



Programs for PORT

1. WAP for 8051 to read data from Port 1 at regular intervals and store the data in R7, R6, R5.

```
MOV    A, #0FFH    ;A=FF hex
MOV    P1,A        ;make P1 an input port
                     ;by writing all 1s to it
MOV    A,P1        ;get data from P1
MOV    R7,A        ;save it in reg R7
ACALL  DELAY        ;wait
MOV    A,P1        ;get another data from P1
MOV    R6,A        ;save it in reg R6
ACALL  DELAY        ;wait
MOV    A,P1        ;get another data from P1
MOV    R5,A        ;save it in reg R5
```

Write a test program for the DS89C420/30 chip to toggle all the bits of P0, P1, and P2 every 1/4 of a second.
Assume a crystal frequency of 11.0592 MHz.

Solution:

;Tested for the DS89C420/30 with XTAL = 11.0592 MHz.

```
ORG    0
BACK:  MOV    A, #55H
        MOV    P0,A
        MOV    P1,A
        MOV    P2,A
        ACALL  QSDelay    ;Quarter of a second delay
        MOV    A, #0AAH
        MOV    P0,A
        MOV    P1,A
        MOV    P2,A
        ACALL  QSDelay
        SJMP  BACK
```

;-----1/4 SECOND DELAY
QSDelay:

```
        MOV    R5, #11
H3:     MOV    R4, #248
H2:     MOV    R3, #255
H1:     DJNZ   R3, H1      ;4 MC for DS89C4x0
        DJNZ   R4, H2
        DJNZ   R5, H3
        RET
        END
```

Delay = $11 \times 248 \times 255 \times 4 \text{ MC} \times 90 \text{ ns} = 250,430 \text{ } \mu\text{s}$

Use an oscilloscope to verify the delay size.

Sub: MDM

Module-4

Sem: IV

Write the following programs.

- (a) Create a square wave of 50% duty cycle on bit 0 of port 1.
- (b) Create a square wave of 66% duty cycle on bit 3 of port 1.

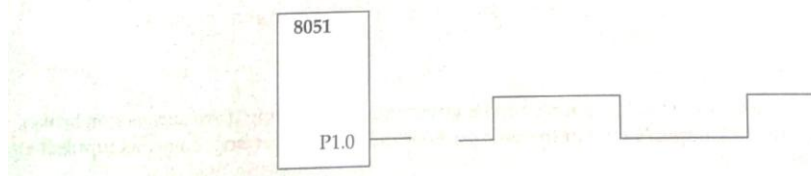
Solution:

- (a) The 50% duty cycle means that the "on" and "off" states (or the high and low portions of the pulse) have the same length. Therefore, we toggle P1.0 with a time delay in between each state.

```
HERE:  SETB  P1.0      ;set to high bit 0 of port 1
        LCALL DELAY    ;call the delay subroutine
        CLR   P1.0      ;P1.0=0
        LCALL DELAY    ;call the delay subroutine
        SJMP  HERE     ;keep doing it
```

Another way to write the above program is:

```
HERE:  CPL   P1.0      ;complement bit 0 of port 1
        LCALL DELAY    ;call the delay subroutine
        SJMP  HERE     ;keep doing it
```



- (b) The 66% duty cycle means the "on" state is twice the "off" state.

```
BACK:  SETB  P1.3      ;set port 1 bit 3 high
        LCALL DELAY    ;call the delay subroutine
        LCALL DELAY    ;call the delay subroutine again
        CLR   P1.3      ;clear bit 2 of port 1 (P1.3=low)
        LCALL DELAY    ;call the delay subroutine
        SJMP  BACK     ;keep doing it
```



Single Bit Instructions

Instruction	Function
SETB bit	Set the bit (bit = 1)
CLR bit	Clear the bit (bit = 0)
CPL bit	Complement the bit (bit = NOT bit)
JB bit,target	Jump to target if bit = 1 (jump if bit)
JNB bit,target	Jump to target if bit = 0 (jump if no bit)
JBC bit,target	Jump to target if bit = 1, clear bit (jump if bit, then clear)



Sub: MDM

Module-4

Sem: IV

Write a program to perform the following

- (a) keep monitoring pin P0.1 until it becomes high
- (b) when P0.1 becomes high, read in the data from port 1
- (c) send a low-to-high pulse on P0.2 to indicate that the data has been read

Solution:

```
SETB P0.1          ;make P0.2 an input
MOV P1,#0FFH       ;make P1 an input port
AGAIN: JNB P0.1, AGAIN ;check if P0.1 is high, if not, keep checking
MOV A, P1          ;if P0.1 is found high, take in data from P1
CLR P0.2           ;clear P0.2
SETB P0.2          ;set P0.2
```

This type of a program is required to read in data from an A to D converter after verifying if the conversion is over (indicated by showing a high on P0.2 here). Note that to make ports or individual port lines to act as input, it is programmed as such by writing 1s to it.

